Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Solving Linear Equations

Name _____ **Date** _____ No calculator **Version 1** Scan to practise this topic online

KEY WORDS equation *(whārite)* · solve *(whakaoti)* · solution *(otinga)* · unknown *(taurangi)*

A One step ●○○

Remember: Do the same thing to both sides. Use the inverse operation: subtract to undo adding, divide to undo multiplying.

1. Solve each equation.

a) $x + 5 = 14 \quad x =$ _____ **b)** $x - 3 = 7 \quad x =$ _____ **c)** $4x = 24 \quad x =$ _____

d) $\frac{x}{3} = 5 \quad x =$ _____ **e)** $x + 12 = 20 \quad x =$ _____ **f)** $x - 9 = -5 \quad x =$ _____

g) $7x = 56 \quad x =$ _____ **h)** $\frac{x}{5} = 6 \quad x =$ _____ **i)** $x + 6 = 4 \quad x =$ _____

j) $3x = -15 \quad x =$ _____ **k)** $x - 2.5 = 3.5 \quad x =$ _____ **l)** $2x = 9 \quad x =$ _____

B Two steps ●●○

Remember: Undo adding and subtracting first. Then undo multiplying and dividing.

2. Solve each equation.

a) $2x + 1 = 9 \quad x =$ _____ **b)** $3x - 2 = 13 \quad x =$ _____ **c)** $5x + 4 = 19 \quad x =$ _____

d) $4x - 3 = 21 \quad x =$ _____ **e)** $2x + 7 = 1 \quad x =$ _____ **f)** $6x - 5 = 7 \quad x =$ _____

g) $\frac{x}{2} + 3 = 7 \quad x =$ _____ **h)** $\frac{x}{4} - 1 = 2 \quad x =$ _____ **i)** $10 - 2x = 4 \quad x =$ _____

j) $3x + 8 = 29 \quad x =$ _____ **k)** $7x - 4 = 31 \quad x =$ _____ **l)** $\frac{x}{5} + 2 = 5 \quad x =$ _____

C The unknown on both sides ●●○

Remember: Collect the x terms on one side and the numbers on the other side.

3. Solve.

a) $3x + 2 = x + 10$ **b)** $5x + 1 = 2x + 10$ **c)** $4x - 3 = 2x + 7$

d) $6x + 5 = 3x - 1$ **e)** $7x - 8 = 5x + 4$ **f)** $2x + 6 = 4x + 2$

g) $3x + 10 = 5x + 2$ **h)** $8x - 1 = 3x + 14$ **i)** $9x + 2 = 4x - 3$

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D Equations with brackets ●●○

> **Remember:** Expand the bracket first, then solve as before.

4. Solve.

a) $2(x+3)=14$ **b)** $3(x-2)=12$ **c)** $4(x+1)=12$

d) $5(x-3)=10$ **e)** $2(2x+1)=14$ **f)** $3(x+4)=9$

g) $6(x-1)=12$ **h)** $2(3x-2)=8$ **i)** $4(x+5)=12$

E Equations with fractions ●●●

> **Remember:** Multiply both sides by the number under the fraction to remove it.

5. Solve.

a) $\dfrac{2x}{3}=6$ **b)** $\dfrac{3x}{5}=6$ **c)** $\dfrac{x+3}{2}=4$

d) $\dfrac{x-4}{3}=2$ **e)** $\dfrac{2x+1}{3}=3$ **f)** $\dfrac{3x-1}{2}=7$

F Problems ●●●

6. Write an equation for each problem, then solve it.

a) I think of a number, multiply it by 3 and add 5. The answer is 23. Find the number.

b) A plumber charges \$15 to come out, plus \$40 for each hour. A job costs \$255. How many hours did it take?

c) A rectangle is x cm wide and $(x+4)$ cm long. Its perimeter is 26 cm. Find x.

d) The angles of a triangle are x°, $2x^\circ$ and 30°. Find x.

★ Extension

7. Solve. Expand both brackets first.

a) $3(x+2)=2x+10$ **b)** $4(x+1)=2(x+5)$ **c)** $5(x-1)=3(x+1)$

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1-1 PRIVATE TUTORS

ALGEBRA

Solving Linear Equations

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS equation *(whārite)* · solve *(whakaoti)* · solution *(otinga)* · unknown *(taurangi)*

A One step ●○○

Remember: Do the same thing to both sides. Use the inverse operation: subtract to undo adding, divide to undo multiplying.

1. Solve each equation.

a) $x + 4 = 11 \quad x =$ _____ **b)** $x - 5 = 6 \quad x =$ _____ **c)** $5x = 35 \quad x =$ _____

d) $\frac{x}{4} = 3 \quad x =$ _____ **e)** $x + 15 = 21 \quad x =$ _____ **f)** $x - 8 = -3 \quad x =$ _____

g) $6x = 54 \quad x =$ _____ **h)** $\frac{x}{3} = 7 \quad x =$ _____ **i)** $x + 7 = 4 \quad x =$ _____

j) $4x = -24 \quad x =$ _____ **k)** $x - 1.5 = 6.5 \quad x =$ _____ **l)** $2x = 7 \quad x =$ _____

B Two steps ●●○

Remember: Undo adding and subtracting first. Then undo multiplying and dividing.

2. Solve each equation.

a) $2x + 3 = 13 \quad x =$ _____ **b)** $3x - 1 = 11 \quad x =$ _____ **c)** $4x + 5 = 17 \quad x =$ _____

d) $5x - 2 = 28 \quad x =$ _____ **e)** $2x + 9 = 1 \quad x =$ _____ **f)** $6x - 7 = 5 \quad x =$ _____

g) $\frac{x}{3} + 2 = 5 \quad x =$ _____ **h)** $\frac{x}{2} - 3 = 4 \quad x =$ _____ **i)** $12 - 3x = 6 \quad x =$ _____

j) $3x + 5 = 32 \quad x =$ _____ **k)** $8x - 3 = 29 \quad x =$ _____ **l)** $\frac{x}{4} + 5 = 8 \quad x =$ _____

C The unknown on both sides ●●○

Remember: Collect the x terms on one side and the numbers on the other side.

3. Solve.

a) $4x + 1 = 2x + 7$ **b)** $5x + 3 = 3x + 11$ **c)** $3x - 5 = x + 7$

d) $7x + 4 = 4x - 5$ **e)** $6x - 7 = 4x + 3$ **f)** $2x + 9 = 5x + 3$

g) $3x + 12 = 6x + 3$ **h)** $9x - 2 = 4x + 8$ **i)** $8x + 5 = 3x - 5$

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D Equations with brackets ●●○

> **Remember:** Expand the bracket first, then solve as before.

4. Solve.

a) $2(x+4)=14$ **b)** $3(x-1)=12$ **c)** $5(x+1)=15$

d) $4(x-2)=16$ **e)** $2(2x+3)=14$ **f)** $3(x+5)=9$

g) $7(x-1)=7$ **h)** $2(3x-1)=16$ **i)** $5(x+4)=15$

E Equations with fractions ●●●

> **Remember:** Multiply both sides by the number under the fraction to remove it.

5. Solve.

a) $\dfrac{2x}{5}=4$ **b)** $\dfrac{3x}{4}=6$ **c)** $\dfrac{x+5}{2}=6$

d) $\dfrac{x-2}{3}=3$ **e)** $\dfrac{2x+1}{5}=3$ **f)** $\dfrac{3x-2}{2}=5$

F Problems ●●●

6. Write an equation for each problem, then solve it.

a) I think of a number, multiply it by 4 and add 3. The answer is 31. Find the number.

b) A plumber charges $20 to come out, plus $60 for each hour. A job costs $320. How many hours did it take?

c) A rectangle is x cm wide and $(x+6)$ cm long. Its perimeter is 32 cm. Find x.

d) The angles of a triangle are x°, $2x^\circ$ and 40°. Find x.

★ Extension

7. Solve. Expand both brackets first.

a) $3(x+4)=2x+17$ **b)** $4(x+2)=2(x+8)$ **c)** $5(x-2)=3(x+2)$

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